



SECI1113: COMPUTATIONAL MATHEMATICS
20232024 – SEMESTER 2

ASSIGNMENT #2

(Submission Date: 12th May 2024)

- 1) Verify the following vectors with axioms 1, 2, 3, 6, 9 and 10

$$(x_1, y_1) + (x_2, y_2) = (x_1x_2, y_1y_2) \quad c(x_1y_1) = (cx_1, cy_1)$$

- 2) a) Let $W = \{(x, y, z) \in R^3 \mid 3x = 2y\}$. Prove that W is subspace of R^3 .
b) Verify W is subspace of $V \in M_{3 \times 2}$. W is the set of all 3×2 matrices of the form of,

$$\begin{bmatrix} a & b \\ a+b & 0 \\ 0 & c \end{bmatrix}$$

- 3) Let $S = \{(4,7,3), (-1,2,6), (2, -3,5)\}$, Is S spanning in R^3 ?
4) Determine the basis and dimension of the solution space of the homogenous system,

$$2x_1 + 2x_2 + 4x_3 - 2x_4 = 0$$

$$x_1 + 2x_2 + x_3 + 2x_4 = 0$$

$$-x_1 + x_2 + 4x_3 - 2x_4 = 0$$

- 5) For the following matrix, calculate the rank, determine the dimension of the column space and provide a basis for the column space:

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 4 \\ 3 & -1 & 2 \end{bmatrix}$$

- 6) Given ordered basis B and B' and given $[v]_B$ find,

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \quad B' = \left\{ \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$
$$[v]_B = \begin{bmatrix} 6 \\ -1 \\ -1 \end{bmatrix}$$

- a) Transition matrix B to B'
b) $[v]_{B'}$

7) Determine whether $T: M_{2,2} \rightarrow R, T(A) = a + b + c + d$, where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a linear transformation.

8) Let $T: R^2 \rightarrow R^2$ be a linear transformation, and let $B = (v_1, v_2)$ is a basis in R^2 ,
If $v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 8 \\ 1 \end{bmatrix}; v_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, T\left(\begin{bmatrix} 4 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 6 \\ -3 \end{bmatrix}$

Find $T\left(\begin{bmatrix} 10 \\ 5 \end{bmatrix}\right)$

9) Let $T: R^5 \rightarrow R^3$ is a linear transformation defined by $T(v) = Av$, where,

$$A = \begin{bmatrix} -1 & 3 & 2 & 1 & 4 \\ 2 & 3 & 5 & 1 & 4 \\ 2 & 0 & 2 & 2 & 0 \end{bmatrix}$$

Find a basis for the null space of T and its dimension

10) Let $T: R^3 \rightarrow R^3$ be a linear operator and $B = \{v_1, v_2, v_3\}$ a basis of R^3 . Suppose

$$T(v_1) = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \quad T(v_2) = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} \quad T(v_3) = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

a) Determine whether $u = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$ is in range of T .

b) Find a basis for $R(T)$.